The Unique Games Theorem: A Short Self-Contained Proof

Ariel Elboim
Independent researcher, ORCID 0009-0001-1341-366X
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Abstract

We give a short proof of the theorem of OpenAI's preprint *The Unique Games Theorem* [1]. It is self-contained apart from three earlier theorems, which are stated as inputs. As in the source, the argument constructs stable nonlinear noise, extracts constant-output certificates, decodes them using only local information, and contradicts parallel repetition on clean coordinates. Relative to the source, several steps are simplified: the noise gadget's recursion is written in direct coordinates, and the survival of linear rank is proved by one induction on the height; genericity is proved by formal differentiation; test acceptance is bounded directly by the measure of good row fibers, without recoloring or concentration; the second player guesses a constant output and Fourier-samples one row; and a triangle of three equations replaces variable cloning. A companion note [7] lists all departures from the source and their credits. A longer presentation that follows the source closely is [6]. The proof is written for a reader comfortable with finite-field linear algebra and Fourier analysis on finite groups.

Status. This is a simplified reconstruction of the proof in [1], prepared with AI agents and checked step by step by separate AI reviewers. These checks do not certify the preprint or verify this paper, and no human expert has reviewed it. OpenAI's repository also contains a Lean 4 formalization of the source theorem, whose Comparator check passed when we ran it; that check concerns OpenAI's formal development, not this paper or its simplifications. Details of the check, and the history of this paper, are in the companion note [7]. This paper is not derived from the formalization. Fortnow's earlier exposition [5] presents the same proof at greater length; this paper was prepared independently of it. The theorem below is the preprint's claim, whose proof we present.

1 The claim and three prior inputs

A unique game labels vertices from a finite alphabet. Each edge carries a permutation: a label at one endpoint specifies exactly one acceptable label at the other. Its value is the largest fraction of constraints satisfied by one labeling.

Theorem 1 (Main claim of [1])**.** *For every fixed $0 < \varepsilon, \delta < 1/2$, there is a deterministic polynomial-time reduction from satisfiability to explicit unweighted unique games on simple bipartite graphs that maps satisfiable formulas to YES games and unsatisfiable formulas to NO games, where*

$$\text{YES} : \operatorname{val}(G) \geq 1 - \varepsilon, \qquad \text{NO} : \operatorname{val}(G) \leq \delta.$$

The alphabet is a fixed binary vector space, and every constraint is a translation. The alphabet and the degree of the running-time polynomial may depend on the two target errors, but not on the input length.

All spaces and linear maps below are over $\mathbb{F}_2$, except where a larger finite field is specified. Uniform samples include zero. Separately specified random choices are independent unless stated otherwise. Intermediate games may have rational weights and repeated constraint occurrences.

Input 1: the parity gap. For every fixed positive rational ξ, Håstad's theorem [2, Theorem 5.4 and its proof] gives a deterministic polynomial-time reduction from satisfiability to lists of three-bit equations, repetitions allowed,

$$x_{v_1} + x_{v_2} + x_{v_3} = b_e, \qquad \text{YES} : \text{OPT} \geq 1 - \xi, \qquad \text{NO} : \text{OPT} \leq \tfrac{1}{2} + \xi,$$

where OPT is the largest satisfiable fraction. The list has one entry per random string of that proof's verifier. If the formula with three literals per clause built in that proof has M clauses, the verifier picks u independent uniform clause–variable pairs, then a fixed budget of b fair bits, padded with unused bits when the chosen clauses share variables; u, the dyadic noise rate and hence b depend only on ξ. So its random strings form a uniform space of size $(3M)^u 2^b$, polynomial in the input, and the list is the unweighted multiset of one equation per string. This size is the number N_0 of entries in Section 2, and gives the common denominators $3N_0$ and $N_{\mathcal{H}}$ of Section 5.

Input 2: Grassmann expansion. The binary Grassmann graph has the ℓ-dimensional subspaces of an ambient space as vertices; adjacency means an intersection of dimension $\ell - 1$. The retention of a nonempty set is the probability that a uniform neighbor of a uniform member stays in the set. For every fixed retention threshold $\zeta > 0$, there are $\alpha > 0$ and an integer $r_{\mathrm{s}} \geq 1$, depending only on ζ, such that every set with retention at least ζ has relative density at least α in an interval

$$\{L : A_0 \subseteq L \subseteq B_0\}, \qquad \dim A_0 + \operatorname{codim} B_0 \leq r_{\mathrm{s}}.$$

This holds for every sufficiently large ℓ, and then every sufficiently large ambient dimension. This is the expansion theorem of Khot–Minzer–Safra [3, Theorem 1.12].

Input 3: projection-game repetition. In a projection game, an answer of the first player determines one required answer of the second. Dinur–Steurer [4, Corollary 1.2] give

$$\operatorname{val}(G) \leq 1 - g \quad \Longrightarrow \quad \operatorname{val}(G^{\otimes n}) \leq (1 - g^2/16)^n. \tag{1}$$

It applies to finite weighted games, lets each player use its entire question tuple, and does not depend on the alphabet. Our games have a fixed projection for each question pair.

Idea and plan. The honest labeling applies one nonlinear map C, and the test perturbs its input by a special noise. The noise rarely changes the output of C, yet every linear family that sees enough of the output notices it with constant probability. Section 3 builds such noise by a race inside a recursive construction. At each level an output change survives with probability $1 - \theta$, so after s levels it survives with probability about $e^{-\theta s}$. For a generic family of linear observations of rank r, the potential H_r drops by at most 3θ per level in expectation. Like a coupon collector, the family therefore needs at least about $H_r/(3\theta)$ levels to die, where $H_r = 1 + \frac{1}{2} + \cdots + \frac{1}{r}$. The harmonic numbers diverge, so a large initial rank wins; the proof takes $s \approx H_{r_0}/(8\theta)$. The algebra needed is that, for a generic family, at most $3r$ elements are aligned. An aligned count of order 2^r, as for arbitrary spaces, would make the waiting times $\sum_j 1/(2^j\theta)$ summable, and this argument would fail. Section 2 builds the translation game and proves its completeness, Section 4 proves its soundness (Proposition 2), and Section 5 removes the weights and amplifies the gap.

2 From parity equations to shared tables

Distinct variables and the base gap. Let $\mathcal{E}_0$ be the list from Input 1, each of its N_0 entries of weight $1/N_0$. Over $\mathbb{F}_2$ a repeated variable cancels, so each entry has three distinct variables or is unary, $x = b$. Replace each unary occurrence of weight w by the three equations $x+z_1+z_2 = b$, $x+z_2+z_3 = b$, $x+z_1+z_3 = b$ of weight $w/3$, with fresh z_1, z_2, z_3. If $x = b$, equal auxiliary bits satisfy all three; if $x \neq b$, all three would need $z_i + z_j = 1$, impossible since the three pairwise sums add to 0 while $1+1+1 = 1$, so at most two hold. Hence every occurrence of weight w, replaced or not, contributes at most $w(\frac{2}{3}+\frac{1}{3}\mathbf{1})$, where $\mathbf{1}$ indicates that the original occurrence is satisfied. Write $\mathcal{E}$ for the processed system with weights ω_e. Extending by zero auxiliary bits gives $\mathrm{OPT}(\mathcal{E}) \geq \mathrm{OPT}(\mathcal{E}_0)$, and summing gives $\mathrm{OPT}(\mathcal{E}) \leq \frac{2}{3} + \frac{1}{3}\,\mathrm{OPT}(\mathcal{E}_0)$, so in the NO case, for $\xi < 1/100$, $\mathrm{OPT}(\mathcal{E}) \leq \frac{5}{6} + \frac{\xi}{3} < 6/7$. In the equation–variable game the first player receives an equation and returns a satisfying triple; the second receives a uniformly chosen one of its three variables and returns a bit. Distinctness makes the pair specify a unique position. The second player's answers form a global assignment. On each equation it violates, every legal triple disagrees with it in some position, which is asked with probability $1/3$; so in the NO case

$$\mathrm{val}(G_{\mathcal{E}}) \leq 1 - \frac{1 - \mathrm{OPT}(\mathcal{E})}{3} \leq 20/21. \tag{2}$$

The noise gadget needed below. Section 3 constructs, for each $p_* > 0$, a threshold r_* such that for every sufficiently large ℓ divisible by 3 there are finite data

$$\begin{gathered} K = \mathbb{F}_2^{\ell} \leq \mathcal{V}, \quad C : \mathcal{V} \to K, \quad a \sim \mu, \\ C(x+h) = C(x) + h \quad (h \in K), \qquad \Pr_{x \text{ uniform},a}[C(x+a) \neq C(x)] \leq p_*, \\ G \leq \mathcal{V}^*,\ \dim(G|_K) \geq r_* \quad \Longrightarrow \quad \Pr_a[\exists g \in G : g(a) \neq 0] \geq 1/8. \end{gathered} \tag{3}$$

The noise law is rational. Crucially, r_* is chosen before ℓ. Fix a linear splitting $\mathcal{V} = K \oplus W$.

Questions and homogeneous answers. Sample $U = (e_1, \ldots, e_k)$ independently with replacement. Put $L_U = \mathbb{F}_2^{1+2k}$, with coordinates $t_0, x_1, y_1, \ldots, x_k, y_k$. The triple for equation e_i is

$$(x_i, y_i, b_{e_i} t_0 + x_i + y_i).$$

Vectors with $t_0 = 1$ are exactly the legal answers. Different positions retain separate answer coordinates, even when their variable identifiers repeat. A projected question O retains either an equation or one of its actual bits in each position; a singleton position records only the variable's identifier, not the equation or the slot it came from. This gives a surjective linear map $\pi : L_U \to L_O$ preserving t_0.

Table keys force sharing. A table is a linear map $P : L_Q \to \mathcal{V}$, or an affine map on the legal answers. On the $t_0 = 0$ part of an equation position, the three answer bits x_i, y_i and $z_i = b_{e_i} t_0 + x_i + y_i$ restrict to exactly the nonzero functionals; a singleton position has only its one bit. So in each position P has rank two, and depends on the whole equation; or rank one, and is v times one answer bit; or vanishes. Hence P factors as $P'\pi'$ through the coarsest projected question that keeps, in each position, exactly what P depends on, omitting the positions where P vanishes. Its key is this factorization: positions, types, identifiers and the map P'. Pulling a projected table back to U and reducing again gives the same key: π only deletes answer bits, so the pullback depends on the same bits, its factorization map is that of the projected table composed with π, and P' is unique. Answer bits are functionals on L_U: at a singleton position keeping the third bit, π keeps $z_i = b_{e_i} t_0 + x_i + y_i$, and $vz_i = v(x_i + y_i) + b_{e_i} v t_0$,

so pulling back carries the intercept automatically and keys need no intercept correction. Such keys therefore already occur among tables on equation tuples.

Take vertices to be key orbits under addition of a constant from K. Choose the representative whose intercept has K-component zero. Restore a table's output by adding its removed constant to the vertex label. Every labeling then defines functions satisfying

$$F_Q(P + he_0^\top) = F_Q(P) + h, \qquad F_U(P'\pi) = F_O(P'). \tag{4}$$

Here $e_0^\top$ is the first-coordinate functional.

The test and completeness. Sample $U \sim \omega^{\otimes k}$, a uniform table P, noise $a \sim \mu$, and a uniform $l \in \mathbb{F}_2^m$, where $m = 1 + 2k$. Require

$$F_U(P) = F_U(P + al^\top). \tag{5}$$

Equality of restored outputs is a translation constraint on the two vertex labels. The sampling probabilities are its rational weights. Call this weighted translation game $\mathcal{H}$. Loops and repeated edges are allowed for now.

Given a global assignment satisfying at least $1 - \xi$ of the equations, answer each retained variable by its global bit, and each retained equation by the global first two bits and the required third bit. Label a key representative by evaluating its table on this answer and applying C. Equivariance makes restoration consistent. With probability at least $1 - k\xi$, every equation in U is satisfied. The answers used for all keys of tables on U are then projections of one legal vector z_U. The compared values are

$$C(Pz_U), \qquad C(Pz_U + a(l^\top z_U)).$$

Since $z_U \neq 0$, the first argument is uniform and $l^\top z_U$ is an independent fair bit. Consequently

$$\mathrm{val}(\mathcal{H}) \geq 1 - k\xi - p_*/2. \tag{6}$$

Why a gadget is needed. Without one ($\mathcal{V} = K$, C the identity, a uniform on K), the labeling above is $F_U(P) = Pz_U$ on tuples whose equations are all satisfied, and $F_U(P + al^\top) = Pz_U + a(l^\top z_U)$. As $l^\top z_U$ is a fair bit independent of a, the test (5) accepts there with probability exactly $\frac{1}{2} + \frac{1}{2|K|}$. The gadget's C absorbs the noise, giving (6), yet every linear family whose restriction to K has large dimension still detects it, which Lemma 3 uses.

3 The latent noise gadget

We prove the contract (3), with r_* depending only on p_*.

3.1 One quadratic block

Take $F = \mathbb{F}_q$, where $q = 2^d$, and put $X = B_{\mathrm{blk}} = F^3$ and $d_{\mathrm{blk}} = 3d$. Identify their binary duals with F^3 using the nondegenerate pairing $\mathrm{Tr}(u \cdot v)$, where $\mathrm{Tr} : F \to \mathbb{F}_2$ is the field trace. Define

$$Q(x) = (x_2x_3, x_1x_3, x_1x_2), \qquad C_{\mathrm{blk}}(x, y) = y + Q(x).$$

For each one-dimensional F-subspace A of F^3 fix $v \in A \setminus \{0\}$, and let

$$U_A = \{(tv, w) : \phi_v(t, w) = 0\}, \qquad \phi_v(t, w) = v \cdot w + t^2 v_1 v_2 v_3 \quad (t \in F,\ w \in F^3).$$

As ϕ_v is binary-linear and onto F, U_A is a binary subspace of dimension $4d - d = d_{\mathrm{blk}}$. One coordinate, $(x_2 + a_2)(x_3 + a_3) + x_2x_3 + a_2a_3 = x_2a_3 + x_3a_2$, gives by symmetry $Q(x + a) + Q(x) +$

$Q(a) = x \times a$, the cross product (signs do not matter in characteristic two). Hence the update (tv, w) changes the output by $w + t^2 Q(v) + t\, x \times v$, whose dot product with v is $\phi_v(t, w)$, as $v \cdot Q(v) = v_1 v_2 v_3$ and $v \cdot (x \times v) = 0$. For a fixed input, each t thus has exactly one w keeping the output, and it lies in U_A. So exactly $q - 1$ of the $q^3 - 1$ nonzero updates keep it, and a uniform nonzero update changes it with probability $1 - \theta$, where $\theta = (q-1)/(q^3-1) = (q^2+q+1)^{-1}$ is one over the number of field lines.

Fix a binary subspace $S \le B^*_{\mathrm{blk}}$ and a linear extension $z \mapsto (\gamma_z, z) \in X^* \oplus B^*_{\mathrm{blk}}$, before choosing A, with $\gamma_z(x) = \mathrm{Tr}(g(z) \cdot x)$ for binary-linear $g : S \to F^3$. For $z \neq 0$,

$$(\gamma_z, z)|_{U_A} = 0 \quad \Longleftrightarrow \quad A = Fz \ \text{ and } \ (z \cdot g(z))^2 = z_1 z_2 z_3. \tag{7}$$

Indeed, the binary functionals vanishing on $\ker \phi_v$ are $\mathrm{Tr}(c\, \phi_v)$, $c \in F$. Matching one with $(t, w) \mapsto \mathrm{Tr}(t\, g(z) \cdot v + z \cdot w)$ forces $z = cv$, so $A = Fz$, and with $t = cs$ leaves $\mathrm{Tr}(s\, g(z) \cdot z + s^2 z_1 z_2 z_3) = 0$ for all $s \in F$. Since $\mathrm{Tr}(s\, g(z) \cdot z) = \mathrm{Tr}(s^2 (g(z) \cdot z)^2)$ and squaring permutes F, nondegeneracy of the trace gives (7).

3.2 Generic spaces work for every extension

Call an r-dimensional binary subspace $S \le F^3$ *generic* if, for every binary-linear $g : S \to F^3$, the alignment equation in (7) has at most $3r$ nonzero solutions. For each fixed r, a uniform r-dimensional subspace is generic with probability tending to one as q tends to infinity.

Use independent uniform parameters $y_{ij} \in F$, and write

$$z_i(t) = \sum_{j=1}^{r} y_{ij} t_j, \qquad c_t = z_1(t) z_2(t) z_3(t), \qquad t \in \mathbb{F}_2^r.$$

Conditional on injectivity of $t \mapsto z(t)$, which fails with probability at most $2^r q^{-3}$, its image is a uniform r-space.

Fix a set $E \subseteq \mathbb{F}_2^r \setminus \{0\}$ with $|E| > 3r$, and form

$$M_{t,(i,j)} = t_j z_i(t)^2 \qquad (t \in E,\ 1 \le i \le 3,\ 1 \le j \le r).$$

Writing $g_i(z(t)) = \sum_j a_{ij} t_j$, the squared alignment equation at t is $\sum_{i,j} M_{t,(i,j)} a_{ij}^2 = c_t$. So an extension aligning on E gives $c = Mu$ with $u_{ij} = a_{ij}^2$; this single system handles every observer.

Over the field $\mathbb{L} = \mathbb{F}_2(y_{ij})$, with the parameters as indeterminates, c is not in the column space $\mathcal{C}$ of M. Indeed, suppose $c = Mu$ with $u \in \mathbb{L}^{3r}$. Differentiate formally in the parameters, extending partial derivatives to $\mathbb{L}$ by the quotient rule. The entries of M involve only squares y_{ij}^2, so in characteristic two every derivative of M vanishes. Hence, for all j, k, l,

$$(t_j t_k t_l)_{t \in E} = \partial_{y_{1j}} \partial_{y_{2k}} \partial_{y_{3l}} c = M\, \partial_{y_{1j}} \partial_{y_{2k}} \partial_{y_{3l}} u \in \mathcal{C}.$$

The three differentiated variables are distinct even when $j = k = l$, so no factorial vanishes; this case gives $t_j^3 = t_j$, so every coordinate vector $(t_j)_{t \in E}$ lies in $\mathcal{C}$. Multiplying a column of M by a coordinate stays in $\mathcal{C}$, since $t_a M_{t,(i,j)} = \sum_h y_{ih}^2\, t_a t_j t_h$ is a combination of the cubic vectors above. Thus $\mathcal{C}$ is closed under coordinatewise multiplication by each t_a, and so contains every positive-degree monomial. The point indicator of a nonzero $e \in E$ vanishes at 0, so it has no constant term, and its values on E, the unit vector at e, lie in $\mathcal{C}$; hence $\dim \mathcal{C} = |E| > 3r$, impossible for $3r$ columns.

So some λ over $\mathbb{L}$ has $\lambda M = 0$ and $\lambda c \neq 0$; scale to polynomials. Parameters admitting $c = Mu$ give $\lambda c = 0$, as $\lambda M = 0$ identically. By Schwartz–Zippel, the nonzero polynomial λc, fixed with E and independent of q, vanishes at uniform F-parameters with probability at most $\deg(\lambda c)/q$. A union over the E of size $3r + 1$, whose number depends only on r, and the conditioning on injectivity, a factor tending to one, finish the proof.

3.3 Recursion reduces the nonlinear error

Each level has input $V_t = W_t \oplus B_{\mathrm{blk}}$ and output map $C_t(w, b) = b + c_t(w)$. Start with $W_0 = 0$, $V_0 = B_{\mathrm{blk}}$ and $C_0(b) = b$. At each later height use one child for each pair in $\mathcal{I} = \{(A, J) : J : B_{\mathrm{blk}} \xrightarrow{\sim} U_A\}$ with binary isomorphisms, and put

$$W_t = W_{t-1}^{\mathcal{I}} \oplus X, \qquad C_t((w_i)_i, x, b) = C_{\mathrm{blk}}\Big((x, b) + \sum_{i \in \mathcal{I}} J_i c_{t-1}(w_i)\Big),$$

which has the required form with $c_t((w_i), x) = C_t((w_i), x, 0)$, so

$$C_t(w, b + h) = C_t(w, b) + h \qquad (h \in B_{\mathrm{blk}}). \tag{8}$$

For child i define the linear injection $\iota_i : V_{t-1} \to V_t$, $\iota_i(u, v) = ((\mathbf{1}_{j=i} u)_j, J_i v)$, with $J_i v \in U_A$ filling (x, b). Let ξ_0 be uniform on B_{blk} and $\xi_t = \iota_i \xi_{t-1}$ for uniform i and independent ξ_{t-1}. Each line has the same number of isomorphisms, so uniform i is a uniform line followed by a fresh uniform isomorphism.

Let $\Delta = v + c_{t-1}(w_i + u) + c_{t-1}(w_i)$ be the output change of the chosen child under noise (u, v); it does not involve the child's last coordinate. For uniform input the child coordinates are uniform and independent of i and the noise, so Δ has i-independent law, nonzero with probability p_{t-1}. Under ξ_t, $\beta = (x, b) + \sum_j J_j c_{t-1}(w_j)$ moves to $\beta + J_i \Delta$, with β uniform conditional on i, child noise and coordinates since (x, b) is uniform and independent of them. Given nonzero Δ and the line, the fresh isomorphism makes $J_i \Delta$ uniform nonzero in U_A. The block count gives $p_t = (1 - \theta) p_{t-1} \le (1 - \theta)^t$, where p_t is the probability that C_t changes under its noise, on a uniform input independent of the noise.

3.4 Linear rank survives the recursion

Put $H_0 = 0$ and $H_r = \sum_{i=1}^{r} 1/i$. Let $G \le V_t^*$, for $t \ge 1$, write (γ_ψ, z_ψ) for the restriction of $\psi \in G$ to $X \oplus B_{\mathrm{blk}}$, and let $S = G|_{B_{\mathrm{blk}}}$ have dimension $r \ge 1$. For child $i = (A, J)$, $(\psi \circ \iota_i)(0, v) = (\gamma_\psi, z_\psi)(Jv)$, so with $E_A = \{(\gamma_\psi, z_\psi)|_{U_A} : \psi \in G\}$, $(\iota_i^* G)|_{B_{\mathrm{blk}}} = J^* E_A$. A linear section $\varsigma : S \to G$ of restriction gives the extension $z \mapsto (\gamma_{\varsigma(z)}, z)$, whose restrictions to U_A lie in E_A. If k_A is the kernel dimension of this restriction map $S \to E_A$, then $\dim E_A \ge r - k_A$. By (7), every nonzero kernel element lies on A and is aligned, and each aligned element lies on exactly one line. Let N_S be the largest number of aligned elements over all extensions; $N_S \le 3r$ if S is generic, and $N_S < 2^r$ always. Then $\sum_A (2^{k_A} - 1) \le N_S$. Since $1/(r - j) \le (j + 1)/r \le 2^j/r$ for $0 \le j < r$, $H_r - H_{r-h} \le (2^h - 1)/r$ for $0 \le h \le r$; averaging over the θ^{-1} lines gives

$$\mathbb{E}_A H_{\dim E_A} \ge H_r - \theta N_S / r. \tag{9}$$

Choose $r_0 \ge 1$ with $e^{1 - H_{r_0}/8} \le p_*$, possible as $H_{r_0} \to \infty$, and set $r_* = r_0$; then let the field degree d be any integer large enough for $2^{r_0}\theta \le \frac{1}{4}$ and nongeneric proportion at most 2^{-r_0} at each rank 1 to r_0. Say G detects ξ if some element is nonzero at ξ. We claim that, for every $t \ge 0$ and every $G \le V_t^*$ with $\dim G \le r_0$, $S = G|_{B_{\mathrm{blk}}}$, $r = \dim S$ ($N_S/r = 0$ if $r = 0$),

$$2 H_{r_0} \Pr[G \text{ detects } \xi_t] \ge H_r - \theta N_S / r - 4\theta t. \tag{10}$$

If $r = 0$ the right side is not positive; if $t = 0$, $G = S$ detects uniform ξ_0 with probability at least $\frac{1}{2}$, and $H_{r_0} \ge H_r$ as $r \le r_0$. Let $t, r \ge 1$. Since $\psi(\xi_t) = (\psi \circ \iota_i)(\xi_{t-1})$, the left side averages over uniform i the left side for $\iota_i^* G$ at height $t - 1$; that family has dimension at most r_0 and restriction $S' = J^* E_A$ of rank $r' = \dim E_A$. Given A, fresh J makes S' uniform at its rank $r' \le r_0$, and $N_{S'}/r' < 2^{r_0}$, so $\mathbb{E}[N_{S'}/r' \mid A] \le 3 + 2^{r_0} 2^{-r_0} = 4$. With (9),

$$\mathbb{E}_i\big[H_{r'} - \theta N_{S'}/r'\big] \ge H_r - \theta N_S / r - 4\theta,$$

so (10) at $t - 1$ for each child gives it at t.

The output. Put $\ell = 3d$, $s = \lfloor H_{r_0}/(8\theta) \rfloor \geq 1$ (as $8\theta \leq 2^{1-r_0} \leq 1$), $\mathcal{V} = V_s$, $W = W_s$, $C = C_s$, and $K = B_{\mathrm{blk}} = \mathbb{F}_2^\ell$ in a fixed binary basis of F^3; let μ be the law of ξ_s. By (8), $C(x+h) = C(x) + h$ for $h \in K$, and $p_s \leq (1-\theta)^s \leq e^{-\theta s} \leq e^{\theta - H_{r_0}/8} \leq p_*$. If $\dim(G|_K) \geq r_0$, the span of r_0 elements of G with independent restrictions to K satisfies (10) with $\theta N_S/r_0 < 2^{r_0}\theta \leq \frac{1}{4}H_{r_0}$ and $4\theta s \leq \frac{1}{2}H_{r_0}$; so it, and hence G, detects ξ_s with probability at least $\frac{1}{8}$. All spaces, maps and noise choices are finite, and uniform choices and their pushforwards have rational probabilities. This proves the contract (3).

4 Soundness

Fix p_* and its gadget threshold r_*. Set $\eta = 2^{-r_*}/16$. Apply Input 2 at retention $\eta/4$ to obtain α, r_{s}. Choose ℓ divisible by 3 and large enough for that input and the gadget, and so that

$$2^{-(\ell - r_{\mathrm{s}})} \leq \alpha/4, \qquad 2^{-\ell} < \eta/4. \tag{11}$$

Now fix the gadget and splitting $\mathcal{V} = K \oplus W$. Only k grows from now on. Write a table as $P = (M, T)$ with values in $K \oplus W$.

The precise experiment. Sample $U \sim \omega^{\otimes k}$. Independently in each position, retain the equation with probability $1 - \beta$, or a uniform one of its variables with probability $\beta = k^{-2/3}$. This gives O and $\pi : L_U \to L_O$. Sample a uniform $A : K \to \mathbb{F}_2^{r_{\mathrm{s}}}$, then independent columns of $T_O : L_O \to W$ and $M_O : L_O \to K$, all uniform except at singleton coordinates: there the column of T_O is zero and that of M_O is uniform in $\ker A$. Put $S_O = AM_O$. The messages are exactly

$$(U, A, T_O\pi, S_O\pi), \qquad (O, A, T_O, S_O). \tag{12}$$

Neither player receives M_O or π. Each may use the fixed function family and independent private randomness, but only its own message.

Proposition 2 (Soundness)**.** *There is k_1, depending only on the fixed auxiliary data, such that* $\mathrm{val}(\mathcal{H}) < 99/100$ *for every integer $k \geq k_1$ and every parity instance with* $\mathrm{OPT}(\mathcal{E}) \leq 6/7$*, as in a NO instance with $\xi < 1/100$.*

Choose k_0 so that every $k \geq k_0$, with $m = 1 + 2k$, gives $2^{-m} \leq \eta/4$ and makes $\ell + m$ exceed Input 2's ambient-dimension threshold at retention $\eta/4$; the condition on v_k is added below. If $k \geq k_0$ and a labeling passes (5) with probability at least 99/100, Sections 4.1–4.2 give two strategies of this kind, returning legal answers (not necessarily efficiently), with

$$\Pr[\pi a_U = a_O] \geq \gamma, \tag{13}$$

where $\gamma > 0$, given in (17), depends only on the fixed auxiliary data. If $\mathrm{OPT}(\mathcal{E}) \leq 6/7$, Section 4.3 bounds the same probability by $\exp(-k^{1/3}/7056)$ for every such pair. Take $k_1 \geq k_0$ with $\exp(-k_1^{1/3}/7056) < \gamma$; for $k \geq k_1$ the two bounds contradict each other, so no labeling passes with probability at least 99/100, that is, $\mathrm{val}(\mathcal{H}) < 99/100$.

4.1 Replacing the noise and extracting constant outputs

For a real function f on a finite binary space, put $\widehat{f}(\phi) = \mathbb{E}_x f(x)(-1)^{\phi(x)}$ for each functional ϕ. A nonzero character averages to zero: translation by a point where it is one reverses its sign. Thus characters form an orthonormal basis, Fourier inversion holds, and Parseval gives $\sum_\phi \widehat{f}(\phi)^2 = \mathbb{E} f^2$.

For fixed (U, T) write $f_{U,T}(M) = F_U(M, T)$ for $M \in \mathrm{Hom}(\mathbb{F}_2^m, K)$. Call a nonempty row fiber $AM = S$, with $A : K \to \mathbb{F}_2^{r_{\mathrm{s}}}$, good if at most r_{s} column conditions $Mq_j = t_j$ cut out a

nonempty slice of it on which one value of $f_{U,T}$ has density at least α; then (U, A, T, S) is good row advice. Let $\mathcal{R}_{U,T}$ be the union of all good row fibers, over all A and S, and write $\|\mathcal{S}\|$ for the fraction of all matrices in a set $\mathcal{S}$.

Lemma 3 (Good row fibers)**.** *For $k \geq k_0$, if a labeling of $\mathcal{H}$ passes (5) with probability at least 99/100, then $\mathbb{E}_{U,T}\|\mathcal{R}_{U,T}\| \geq \eta$.*

Proof. Noise comparison. Fix U. For a noise law ν on $\mathcal{V}$ let $\mathrm{acc}_\nu = \Pr[F_U(P) = F_U(P + al^\top)]$, with P and l uniform and $a \sim \nu$, and let ν_K be uniform on K. For $u \in K$ put $f_u = \mathbf{1}_{F_U = u}$, so that $\mathbf{1}_{F_U(P)=F_U(P')} = \sum_u f_u(P) f_u(P')$. Expand each f_u in characters $g = (g_1, \ldots, g_m) \in (\mathcal{V}^*)^m$ of the table space. Orthogonality over P and averaging over l give

$$\mathrm{acc}_\nu = \sum_g w(g)\lambda_\nu(g), \qquad w(g) = \sum_u \widehat{f_u}(g)^2, \qquad \lambda_\nu(g) = \Pr_{a\sim\nu}[g_i(a) = 0 \text{ for all } i] \geq 0,$$

and Parseval makes the weights sum to $\sum_u \mathbb{E} f_u = 1$. Call g low when the restrictions $g_i|_K$ span dimension less than r_*, and let m_{low} be the mass on low frequencies. For the others, the family $\mathrm{span}\{g_i\}$ has restriction to K of dimension at least r_*, and some element of it is nonzero at a exactly when some g_i is. So detection gives $\lambda_\mu \leq 7/8$, whence $\mathrm{acc}_\mu \leq m_{\mathrm{low}} + \frac{7}{8}(1 - m_{\mathrm{low}})$. On the low frequencies $\lambda_{\nu_K} = 2^{-\dim \mathrm{span}(g_i|_K)} \geq 2^{-r_*}$, so $\mathrm{acc}_{\nu_K} \geq 2^{-r_*} m_{\mathrm{low}} \geq 2^{-r_*}(8\,\mathrm{acc}_\mu - 7)$. The noise ν_K leaves T fixed, so $\mathrm{acc}_{\nu_K} = \mathbb{E}_T \mathrm{acc}(f_{U,T})$, where $\mathrm{acc}(f) = \Pr[f(M) = f(M + al^\top)]$ for M, $a \in K$ and l uniform. The bound is linear in acc_μ, whose average over U is at least 99/100, so $\mathbb{E}_{U,T}\,\mathrm{acc}(f_{U,T}) \geq 0.92 \cdot 2^{-r_*} = 14.72\eta > 3\eta$.

Covering. It remains to show $\mathrm{acc}(f) \leq \eta + 2\|\mathcal{R}\|$ for each (U, T), with $f = f_{U,T}$ and $\mathcal{R} = \mathcal{R}_{U,T}$, since averaging then gives $\mathbb{E}_{U,T}\|\mathcal{R}_{U,T}\| \geq (3\eta - \eta)/2 = \eta$. The *step retention* $\mathrm{ret}(\mathcal{S})$ of a nonempty set $\mathcal{S}$ is the probability that the uniform rank-one step $M \mapsto M + al^\top$, zero step included, keeps a uniform member inside it. For each value u put $\mathcal{S}_u = f^{-1}(u) \setminus \mathcal{R}$, and suppose that some nonempty $\mathcal{S}_u$ has $\mathrm{ret}(\mathcal{S}_u) \geq \eta$. The choice of ℓ and k_0 lets Input 2 apply at retention $\eta/4$ and makes the step zero with probability $2^{-\ell} + 2^{-m} - 2^{-\ell-m} \leq \eta/2$, so conditioning on nonzero factors leaves retention at least $\eta/2$. Identify M with its graph

$$\Gamma_M = \{(x, M^\top x) : x \in \mathbb{F}_2^\ell\}, \qquad \dim(\Gamma_M \cap \Gamma_N) = \ell - \mathrm{rank}(M - N).$$

These form the matrix chart in the Grassmann graph on $\mathbb{F}_2^\ell \oplus \mathbb{F}_2^m$. A chart vertex has $(2^\ell - 1)(2^m - 1)$ chart neighbors and $(2^\ell - 1)(2^{m+1} - 2)$ neighbors in total. For the latter count, choose a hyperplane in the vertex and a different one-dimensional extension in the quotient. For the former, use the unique nonzero rank-one factorization over $\mathbb{F}_2$. Exactly half the neighbors are in the chart, so the lifted $\mathcal{S}_u$ has Grassmann retention at least $\eta/4$. Input 2 gives it relative density at least α in an interval, hence also in the interval's intersection with the chart, since the chart contains the whole lifted set. A basis (d_i, s_i) of A_0 and a basis (t_j, q_j) of $B_0^\perp$, for the dot product under which $\Gamma_M^\perp = \{(Mv, v)\}$, express this intersection as the nonempty slice

$$\{M : d_i^\top M = s_i^\top,\ Mq_j = t_j\}, \qquad i \leq n_{\mathrm{row}},\ j \leq n_c,\ n_{\mathrm{row}} + n_c \leq r_{\mathrm{s}}.$$

Its row conditions say $A_* M = S_*$, where A_* has the rows $d_i^\top$ padded with zeros. As $\mathcal{S}_u \subseteq f^{-1}(u)$, this row fiber is good and lies in $\mathcal{R}$; yet the slice meets $\mathcal{S}_u$, which is disjoint from $\mathcal{R}$. Hence every nonempty $\mathcal{S}_u$ has step retention below η.

Let $M' = M + al^\top$ be the step from a uniform M; both endpoints are uniform. An accepted step with neither endpoint in $\mathcal{R}$ stays in one set $\mathcal{S}_u$, so its probability is $\sum_u \|\mathcal{S}_u\| \,\mathrm{ret}(\mathcal{S}_u) \leq \eta$, the sum over nonempty sets. A step with an endpoint in $\mathcal{R}$ has probability at most $2\|\mathcal{R}\|$. □

For each matrix in $\mathcal{R}_{U,T}$ of Lemma 3, at least one of the $2^{\ell r_{\mathrm{s}}}$ possible row maps witnesses goodness. Hence, under a uniform full table and independent uniform A,

$$\Pr[(U, A, T, AM) \text{ is good row advice}] \geq \eta\, 2^{-\ell r_{\mathrm{s}}}. \tag{14}$$

4.2 The private decoders and their agreement

Fix U and A, and put $n = |\mathcal{V}|$. Compare the law $\mathcal{L}_{\mathrm{pad}}$ of the padded full table $(M_O\pi, T_O\pi)$, with its intercept and averaged over the projection choices, with the law $\mathcal{L}_{\mathrm{unif}}$ of a uniform table on L_U. Blocks are independent, and a retained equation has uniform slopes. A singleton block has some slope law Q_A, and $Q_A \le n^2 P_0$ for the uniform law P_0 on $\mathcal{V}^2$, so $\chi^2(Q_A\|P_0) := \mathbb{E}_{P_0}(Q_A/P_0 - 1)^2 = \mathbb{E}_{Q_A}(Q_A/P_0) - 1 < n^2$. Mixing with weight β multiplies the centered ratio by β, and $1 + \chi^2$ multiplies over independent blocks. The intercept changes nothing: pulling back adds b_{e_i} times each third-bit singleton column to it, and an independent uniform intercept stays uniform after adding terms that depend on the slopes. As Cauchy–Schwarz bounds total variation by half the square root of χ^2,

$$\|\mathcal{L}_{\mathrm{pad}} - \mathcal{L}_{\mathrm{unif}}\|_{\mathrm{TV}} \le v_k := \frac{1}{2}\sqrt{\left(1 + n^2\beta^2\right)^k - 1}. \tag{15}$$

Averaging over A, the same bound compares the joint law of A and the padded table with an independent uniform pair. It does *not* compare joint laws that include π. Put $H = \ker A$. By (11), $|H| \ge 2^{\ell - r_{\mathrm{s}}} \ge 4/\alpha$, so $1/(|H|-1) \le 2/|H| \le \alpha/2$. Put $b_* := \eta\, 2^{-2\ell r_{\mathrm{s}}}$. As $k\beta^2 = k^{-1/3} \to 0$, enlarge k_0 so that $v_k \le \alpha b_*/8$ for every $k \ge k_0$.

The strategies. Select a certificate deterministically from each good advice, using a fixed order and the function family. Put $Z = \mathrm{span}\{q_j\}$ and

$$\mathcal{A} = Z \cap \{v : e_0^\top v = 1\}, \qquad |\mathcal{A}| \le 2^{r_{\mathrm{s}}}. \tag{16}$$

On its message $(U, A, T_O\pi, S_O\pi)$, the first player samples uniformly from this list, and gives any legal answer when the list is empty or the advice is not good. On its message (O, A, T_O, S_O), the second player samples a uniform guess $u \in K$, a uniform $h \in H \setminus \{0\}$ and a uniform $B \in \mathrm{Hom}(L_O, H)$. It fixes M_0 with $AM_0 = S_O$ and, with $h\lambda : x \mapsto \lambda(x)h$, puts

$$f(N) = \mathbf{1}_{F_O(M_0+N,T_O)=u} - \mathbf{1}_{F_O(M_0+N,T_O)=u+h}, \qquad \varphi(\lambda) = f(B + h\lambda)$$

for $N \in \mathrm{Hom}(L_O, H)$ and $\lambda \in L_O^*$. Indexing characters of L_O^* by $z \in L_O$, it returns $a_O = z$ with probability $\widehat{\varphi}(z)^2/\mathbb{E}\varphi^2$, a law by Parseval, or a fixed legal answer if $\varphi = 0$. Folding (4) gives $f(N + he_0^\top) = -f(N)$; substituting $\lambda + e_0^\top$ then gives $\widehat{\varphi}(z) = -(-1)^{e_0^\top z}\widehat{\varphi}(z)$, so a_O is legal. Each player uses only its own message and private randomness; u_R, J, Z', X, $\tilde{N}$ and Y below enter only the analysis.

Agreement. Expose $D = (U, O, \pi, A, T_O, S_O, h)$. Given D, both messages are fixed, so the first player's list choice, $N = M_O - M_0$ and the decoder's (u, B, z) are independent; N is uniform on $\mathrm{Hom}(L_O, H)$, since at singleton coordinates too the row fiber is $\ker A$. The first player's full row advice $R = (U, A, T_O\pi, S_O\pi)$ is a function of D, and so is its certificate. For good R, let u_R be the first value, in the fixed order, of density at least α on its chosen column slice. Let J mean that R is good and its column conditions $M_O\pi q_j = t_j$ hold. They read $N\pi q_j = t_j - M_0\pi q_j$, so J is a function of D and $N|_{Z'}$, where $Z' = \pi Z$. Put $X = \mathbb{E}[f(N) \mid D, N|_{Z'}]$ for $u = u_R$ if R is good, and $X = 0$ otherwise.

If R is not good, $X = 0$ and the bound below is trivial. Otherwise, given D, the guess $u = u_R$ is right with probability $2^{-\ell}$; assume it. Take λ uniform on L_O^*, independent of D and B, and put $\tilde{N} = B + h\lambda$, so that $\varphi(\lambda) = f(\tilde{N})$ and $\tilde{N}|_{Z'}$ is a function of B and $\lambda|_{Z'}$. Averaging φ over each set $\lambda|_{Z'} = \mathrm{const}$ keeps exactly its coefficients at Z', so Parseval and $\mathbb{E}\varphi^2 \le 1$ give, even if $\varphi = 0$,

$$\Pr[a_O \in Z' \mid D, u, B] \ge \sum_{z \in Z'} \widehat{\varphi}(z)^2 = \mathbb{E}[Y^2 \mid D, B], \qquad Y := \mathbb{E}[f(\tilde{N}) \mid D, B, \lambda|_{Z'}].$$

By the tower property and conditional Jensen,

$$\mathbb{E}[Y^2 \mid D] \geq \mathbb{E}\big[\mathbb{E}[f(\tilde{N}) \mid D, \tilde{N}|_{Z'}]^2 \mid D\big] = \mathbb{E}[X^2 \mid D],$$

the equality because, given D, the uniform B makes $\tilde{N}$ uniform on $\mathrm{Hom}(L_O, H)$, like N, and f is fixed by D. Averaging over B gives $\Pr[a_O \in Z',\ u = u_R \mid D] \geq 2^{-\ell}\mathbb{E}[X^2 \mid D]$. On this event $a_O = \pi z'$ for some $z' \in Z$, and $z' \in \mathcal{A}$, as a_O is legal and π preserves the first coordinate. The first player's independent uniform choice from $\mathcal{A}$ hits z' with probability at least 2^{-r_s}, so $\Pr[\pi a_U = a_O \mid D] \geq c\,\mathbb{E}[X^2 \mid D]$ with $c := 2^{-r_\mathrm{s}-\ell}$.

It remains to bound $\mathbb{E}[\mathbf{1}_J X] = \mathbb{E}[\mathbf{1}_J f(N)]$, with $u = u_R$. For the full table $(M, T) = (M_O\pi, T_O\pi)$, sharing gives $f(N) = \mathbf{1}_{F_U(M,T)=u_R} - \mathbf{1}_{F_U(M,T)=u_R+h}$, and J says that the advice (U, A, T, AM) is good and $Mq_j = t_j$. So the score $\mathbf{1}_J f(N)$ is one formula, bounded by one, in U, A, h and the full table, not involving π; as h depends only on A, (15) applies to it. Under the uniform law, given good advice R, M is uniform on its row fiber, each nonempty column slice imposes at most ℓr_s additional scalar equations on it, and h is independent of M; so the formula has conditional mean at least $\alpha - 1/(|H| - 1) \geq \alpha/2$ given R and J; and, as the slice has probability at least $2^{-\ell r_\mathrm{s}}$ in its fiber, (14) gives $\Pr_{\mathrm{unif}}(J) \geq b_*$. Hence $\mathbb{E}[\mathbf{1}_J f(N)] \geq \alpha b_*/2 - 2v_k \geq \alpha b_*/4$, and Cauchy–Schwarz gives

$$\Pr[\pi a_U = a_O] \geq c\,\mathbb{E}X^2 \geq c\big(\mathbb{E}[\mathbf{1}_J X]\big)^2 \geq \frac{c\alpha^2 b_*^2}{16} = \frac{\alpha^2\eta^2}{16}\,2^{-4\ell r_\mathrm{s} - r_\mathrm{s} - \ell} =: \gamma. \tag{17}$$

This proves (13) for the original unconditioned experiment.

4.3 Clean coordinates force disagreement

Let $\mathrm{OPT}(\mathcal{E}) \leq 6/7$, and take any two strategies that use only their own messages (12) and private randomness and return legal answers. By (2) the base game has value at most $20/21$. Call the singleton positions clean, and let I be their set. Given A, positions are independent and the law of a position's columns depends only on whether it is a singleton. So the experiment can be sampled in two stages: first A, I, all T_O, M_O columns and equations outside I; then for each $i \in I$ an independent base round $e_i \sim \omega$ with a uniform variable. At clean positions T_O, S_O columns vanish, so the pulled-back table does not depend on that position. Given the first stage, the first message is a function of the clean equations and the second of the clean variables, with one answer coordinate per position even when identifiers repeat. With the first stage as a fixed parameter, sent to neither player, and keeping their clean answers, the strategies thus play the $|I|$-fold base game legally, full agreement winning every round. Mixing over private randomness, Input 3 gives, also for $I = \varnothing$,

$$\Pr[\pi a_U = a_O \mid \text{first stage}] \leq (1 - 1/7056)^{|I|}.$$

Since $|I| \sim \mathrm{Bin}(k, \beta)$,

$$\Pr[\pi a_U = a_O] \leq (1 - \beta/7056)^k \leq \exp(-k^{1/3}/7056) \longrightarrow 0. \tag{18}$$

This bounds the same *unconditioned* experiment as (13) and proves Proposition 2.

5 Removing weights, amplifying, and writing the output

An unweighted simple bipartite graph. Weights of $\mathcal{E}$ are multiples of $1/(3N_0)$ and μ has fixed denominator D_μ, so all $\mathcal{H}$ probabilities are multiples of $1/N_\mathcal{H}$ with $N_\mathcal{H} = (3N_0)^k|\mathcal{V}|^m 2^m D_\mu$ polynomial (only N_0 grows). Replace a constraint of weight w by $wN_\mathcal{H}$ four-edge paths with three fresh internal vertices: three identity permutations then the original constraint. For fixed

endpoint labels, all four edges can be satisfied exactly when the original constraint holds, and otherwise exactly three; disjoint interiors let optima combine. Hence for G_{bip},

$$\mathrm{val}(G_{\mathrm{bip}}) = 1 - \frac{1 - \mathrm{val}(\mathcal{H})}{4}. \tag{19}$$

Place original vertices and path midpoints on one side and the other internal vertices on the other; loops become four-cycles, parallel copies get disjoint interiors, so the graph is simple and bipartite. Reversing an edge preserves binary translations.

Parameter order. Choose positive rationals $\varepsilon_0 \le \varepsilon$, $\delta_0 \le \delta$. By Proposition 2 and (19), the NO gap after subdivision is at least $1/400$. Since (1) is alphabet-independent, first fix t with

$$\left(1 - \frac{1}{2\,560\,000}\right)^t \le \delta_0, \qquad 2\,560\,000 = 16 \cdot 400^2. \tag{20}$$

Next choose a positive rational $p_* \le \varepsilon_0/t$; then r_*, η, α, r_{s}, ℓ and the gadget as in Section 4; then $k = k_1$ of Proposition 2; finally a positive rational $\xi < \min\{1/100, \varepsilon_0/(2kt)\}$. All are uniform over the instance and the function family, hardcoded into one machine for fixed targets, so their existence suffices; no uniform computability in targets is claimed.

The final game. Output $G = G_{\mathrm{bip}}^{\otimes t}$. In the YES case (6) gives $\mathrm{val}(\mathcal{H}) \ge 1 - \mathrm{err}$ with $\mathrm{err} = k\xi + p_*/2 \le \varepsilon_0/t$; a good labeling in every coordinate and a union bound give

$$\mathrm{val}(G) \ge 1 - t\,\mathrm{err}/4 \ge 1 - \varepsilon_0/4 \ge 1 - \varepsilon.$$

In the NO case, (1) and (20) give $\mathrm{val}(G) \le (1 - 1/2\,560\,000)^t \le \delta_0 \le \delta$. A pair of product vertices specifies at most one edge per coordinate, so G is simple. It is bipartite, and independent uniform base edges give uniform product edges. Its constraints are translations over the fixed alphabet $K^t = \mathbb{F}_2^{\ell t}$. Keys need only fixed-dimensional linear algebra, even comparing every pair of keys is polynomial, and t is fixed, so G is explicit in polynomial time, proving Theorem 1.

Departures from the source proof [1], credits for each simplification, the review history and the Lean check are listed in the companion note [7], deposited with this paper.

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[1] OpenAI. *The Unique Games Theorem.* Preprint, 23 September 2026, 58 pages. Pinned source revision: GitHub source and paper.

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